

Title: Supporting Information for The Frequency and Topology of Pseudoorthologs

Authors: Megan L. Smith¹ and Matthew W. Hahn¹

¹Department of Biology and Department of Computer Science, Indiana University, Bloomington,
IN 47405, USA

Corresponding Author: Megan L. Smith; email: mls16@indiana.edu)

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Supporting Tables S1-S5

Supporting Text—The four-taxon case

To explore the effects of adding an extra taxon to branch t_1 , we considered the species tree $((((A,B),C),D))$. We calculated the probabilities of orthologs, concordant pseudoorthologs, and each of the possible 14 discordant pseudoorthologs (Supporting Fig. S11) in this case. As expected, the probability of orthologs and concordant pseudoorthologs was reduced in the four-taxon case (Supporting Fig. S12). Adding the fourth taxon led to higher ratios of concordant to discordant topologies than observed in the three-taxon case, with the exception of the ‘Class 5’ and ‘Class 6’ (Supporting Fig. S12) pseudoorthologs. The ratio of the concordant topologies to these two discordant topologies was very similar to the concordant to discordant ratios in the three-taxon case, although it became slightly lower as the length of branch t_1 increased (Supporting Figure S12). Notably, these two discordant pseudoorthologs are concordant with respect to the relationships among the original three taxa (A, B, and C; Supporting Fig. S11). The probability of any discordant topology never exceeded that of the concordant topology (Supporting Fig. S12).

Supporting Figures

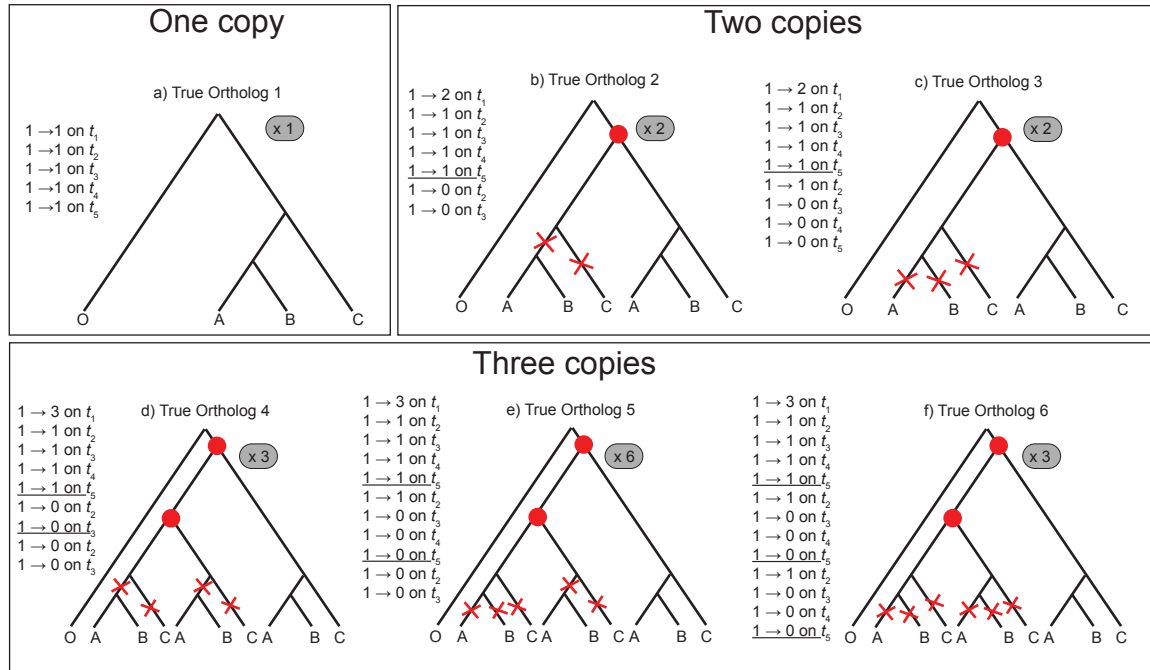


Fig. S1. Orthologs considered. Red dots indicate duplication events, and red Xs indicate loss events. The transitions necessary to observe each configuration are listed on the side of each panel. The number of copies at the bottom of branch t_1 are one (panel a), two (panels b, c), or three (panels d, e, f). The 'x 2' or 'x 6' indicates the number of potential arrangements considered for each scenario. For example, for true ortholog 2, the losses could occur in either one of the two copies.

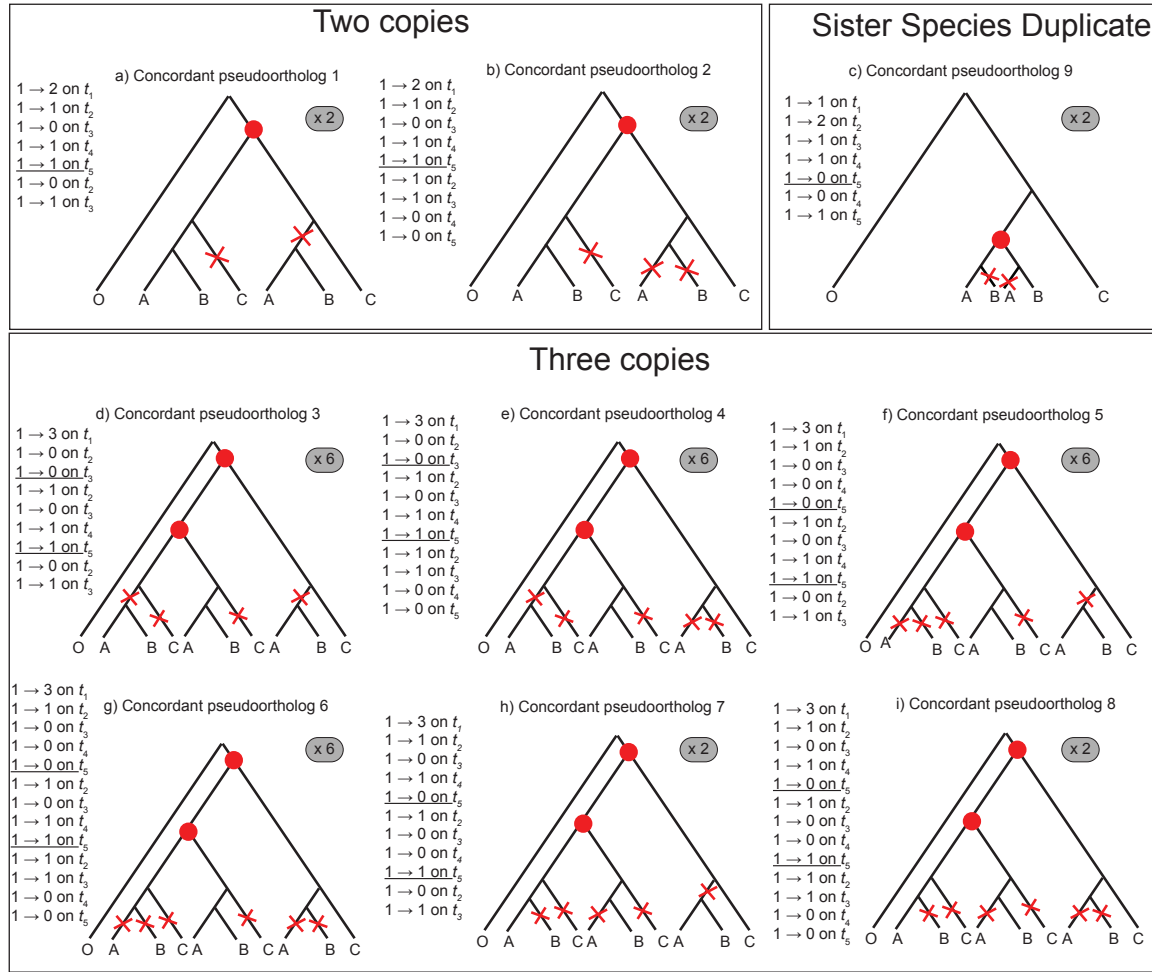
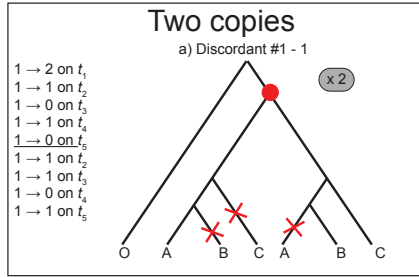


Fig. S2. Concordant pseudoortholog scenarios considered. Red dots indicate duplication events, and red Xs indicate loss events. The transitions necessary to observe each configuration are listed on the side of each panel. The number of copies at the bottom of branch t_1 are one (panel c), two (panels a, b), or three (panels d-i). The 'x 2' or 'x 6' indicates the number of potential arrangements considered for each scenario.

Discordant Topology 1



Discordant Topology 2

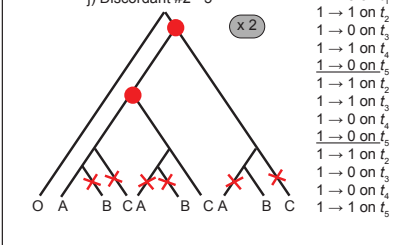
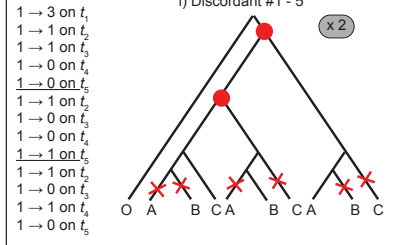
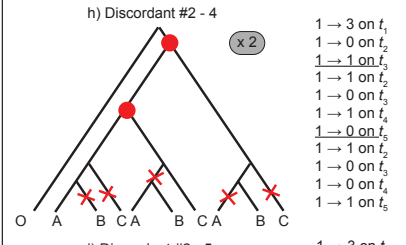
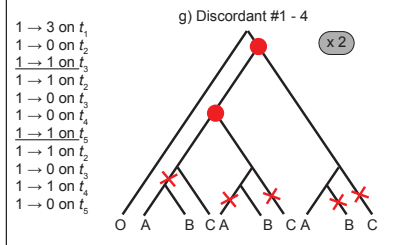
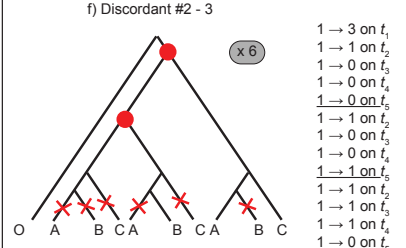
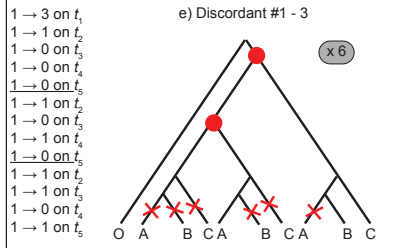
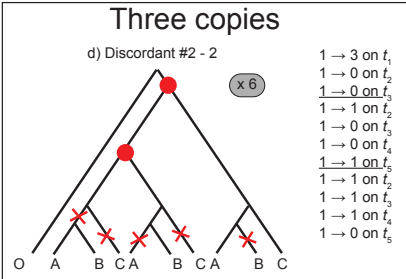
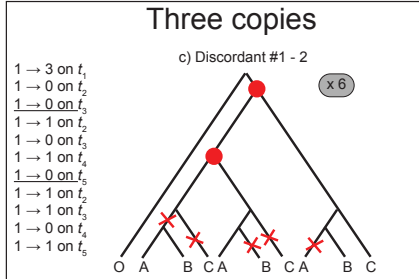
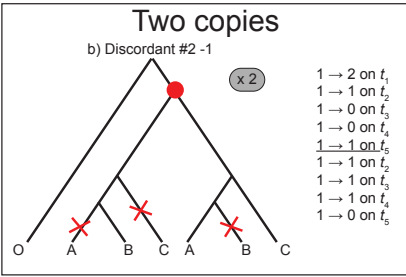


Fig S3. Discordant pseudoortholog scenarios considered. Red dots indicate duplication events, and red Xs indicate loss events. The transitions necessary to observe each configuration are listed on the side of each panel. The number of copies at the bottom of branch t_1 are two (panels a, b) or three (panels c-j). The 'x 2' or 'x 6' indicates the number of potential arrangements considered for each scenario, with the two discordant topologies considered separately in each column.

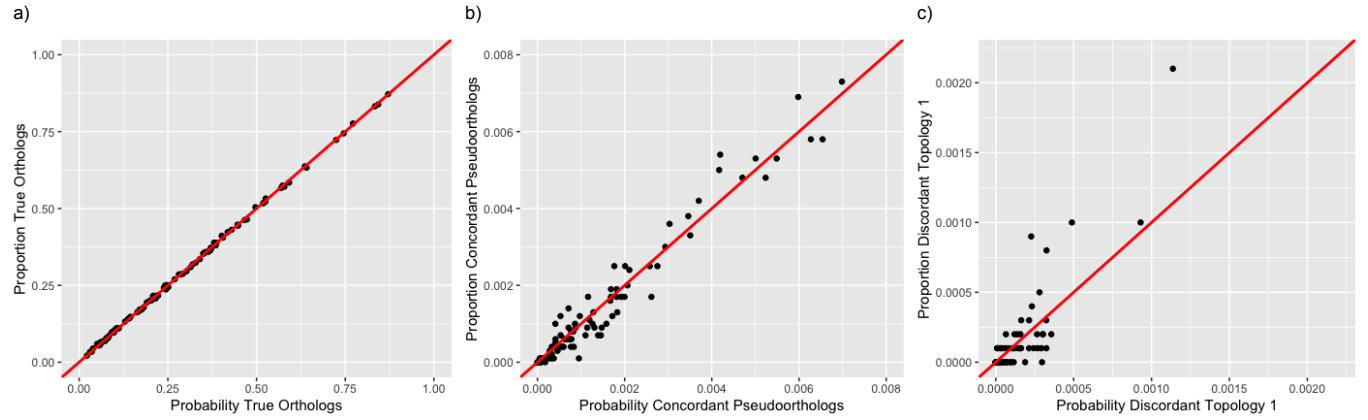


Fig S4. Results of simulations. Red lines indicate the predicted 1:1 relationship between expected probabilities and observed probabilities from simulations. a) The proportion of true orthologs in simulations versus the probability of orthologs under the model. b) The proportion of concordant pseudoorthologs versus the probability of concordant pseudoorthologs under the model. c) The proportion of discordant pseudoortholog 1 versus the probability of discordant pseudoortholog 1 under the model. Note that there are few events in this last category, as the probability of discordant pseudoorthologs is always very low.

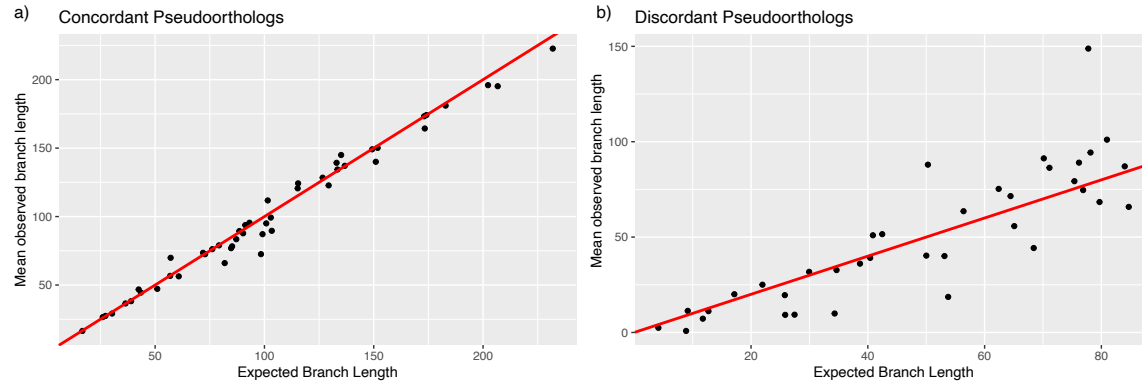


Fig S5. Results of simulations to evaluate branch length predictions. Red lines indicate the predicted 1:1 relationship between the expected internal branch lengths and the observed internal branch lengths in simulated datasets. Each datapoint is the average internal branch length of all simulations under a set of parameters that match the relevant scenario. For each set of parameters, we conducted 10,000 simulations. a) The expected versus observed internal branch lengths of concordant pseudoortholog 1 (Figure 1d; Figure S2a). b) The expected versus observed internal branch lengths of discordant pseudoortholog 1 when there were two copies at the end of branch t_1 (Figure 1e; Figure S3a).

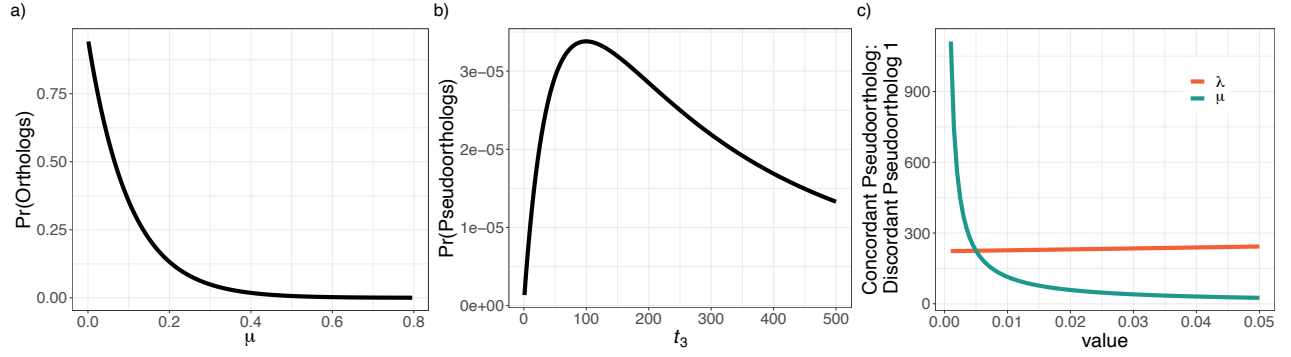


Figure S6. a) The absolute probability of orthologs at values of μ ranging from 0.001 to 0.8 ($t_1=5$, $t_2=1$, $t_3=2$, $t_4=1$, $t_5=1$, $\lambda=0.005$). Here, we show only the results of varying μ because the results of varying λ are identical. b) The absolute probability of pseudoorthologs at values of t_3 ranging from 1 to 500 ($\lambda=0.005$, $\mu=0.005$, $t_1=5$, $t_2=1$, $t_4=1$, $t_5=1$). c) The ratio of the concordant pseudoortholog to one discordant pseudoortholog at values of λ and μ ranging from 0.001 to 0.05 ($t_1=5$, $t_2=1$, $t_3=2$, $t_4=1$, $t_5=1$). The green line shows the ratio of the concordant pseudoortholog to one discordant pseudoortholog when μ varies and λ is held constant at 0.005, while the orange line shows the ratio of the concordant pseudoortholog to one discordant pseudoortholog when λ varies and μ is held constant at 0.005.

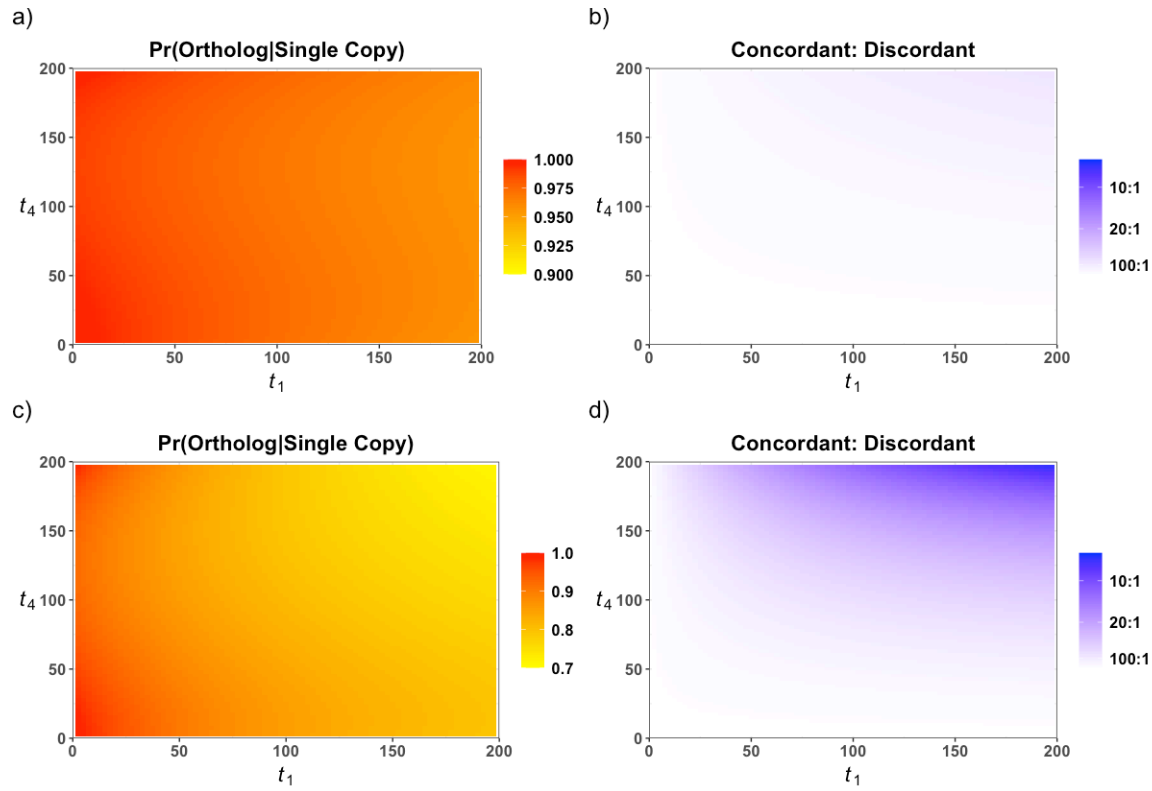


Fig S7. Conditional probabilities of orthologs, pseudoorthologs, and discordance. Identical to main Figure 3, except scales for a and c are modified to facilitate fine-scale visualization of variation in probabilities. Here, branch length $t_3 = 198.9$ mya. Branch length t_1 varies from 0.0001 to 200 mya, while branch length t_4 varies from 0.0001 to 198.8 mya; branch length t_2 is constrained such that the sum of t_2 and t_4 equals t_3 . a) The conditional probability of orthologs given that a single copy is present in each species, with moderate rates of duplication and loss ($\lambda=0.002$ per my, $\mu=0.002$ per my). b) The ratio of the concordant topology (orthologs and concordant pseudoorthologs) to one discordant topology given that a single copy is present in each species, with moderate rates of duplication and loss ($\lambda=0.002$, $\mu=0.002$). c) The conditional probability of orthologs given that a single copy is present in each species with high rates of duplication and loss ($\lambda=0.005$, $\mu=0.005$). d) The ratio of the concordant topology (orthologs and concordant pseudoorthologs) to one discordant topology given that a single copy is present in each species with high rates of duplication and loss ($\lambda=0.005$, $\mu=0.005$).

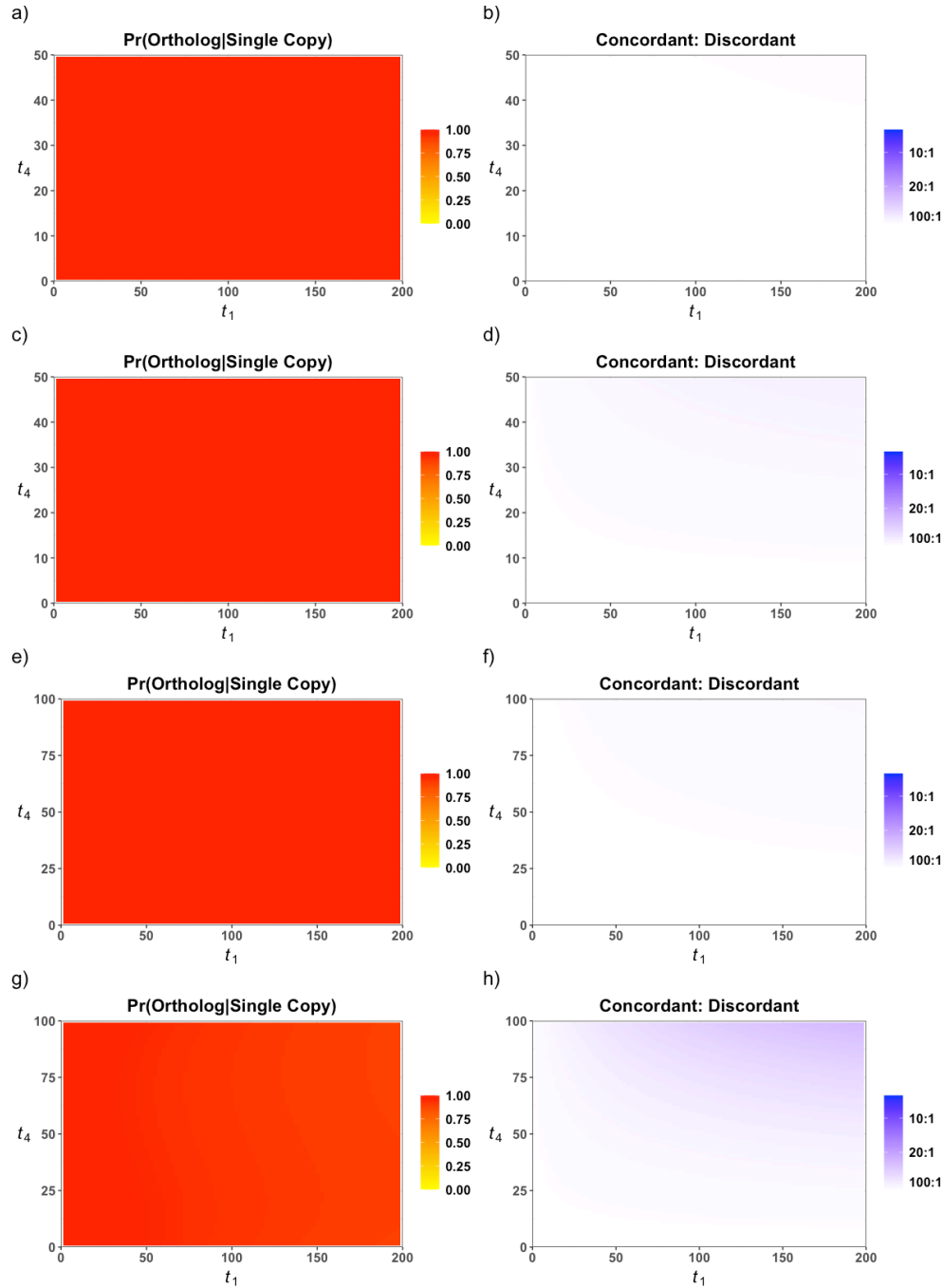


Fig S8. Probabilities of orthologs, pseudoorthologs, and discordance conditional on a single copy per species. Branch length t_1 varies from 0.0001 to 200 mya, while branch length t_4 varies from 0.0001 to $t_3 - 0.1$; branch length t_2 is constrained such that the sum of t_2 and t_4 equals t_3 . a) The conditional probability of orthologs ($\lambda=0.002$ per my, $\mu=0.002$ per my, $t_3=50$). b) The ratio of the concordant topology to one discordant topology ($\lambda=0.002$, $\mu=0.002$, $t_3=50$). c) The conditional probability of orthologs ($\lambda=0.005$, $\mu=0.005$, $t_3=50$). d) The ratio of the concordant topology to one discordant topology ($\lambda=0.005$, $\mu=0.005$, $t_3=50$). e) The conditional probability of orthologs ($\lambda=0.002$, $\mu=0.002$, $t_3=100$). f) The ratio of the concordant topology to one discordant topology ($\lambda=0.002$, $\mu=0.002$, $t_3=100$). g) The conditional probability of orthologs ($\lambda=0.005$, $\mu=0.005$, $t_3=100$). h) The ratio of the concordant topology to one discordant topology ($\lambda=0.005$, $\mu=0.005$, $t_3=100$).

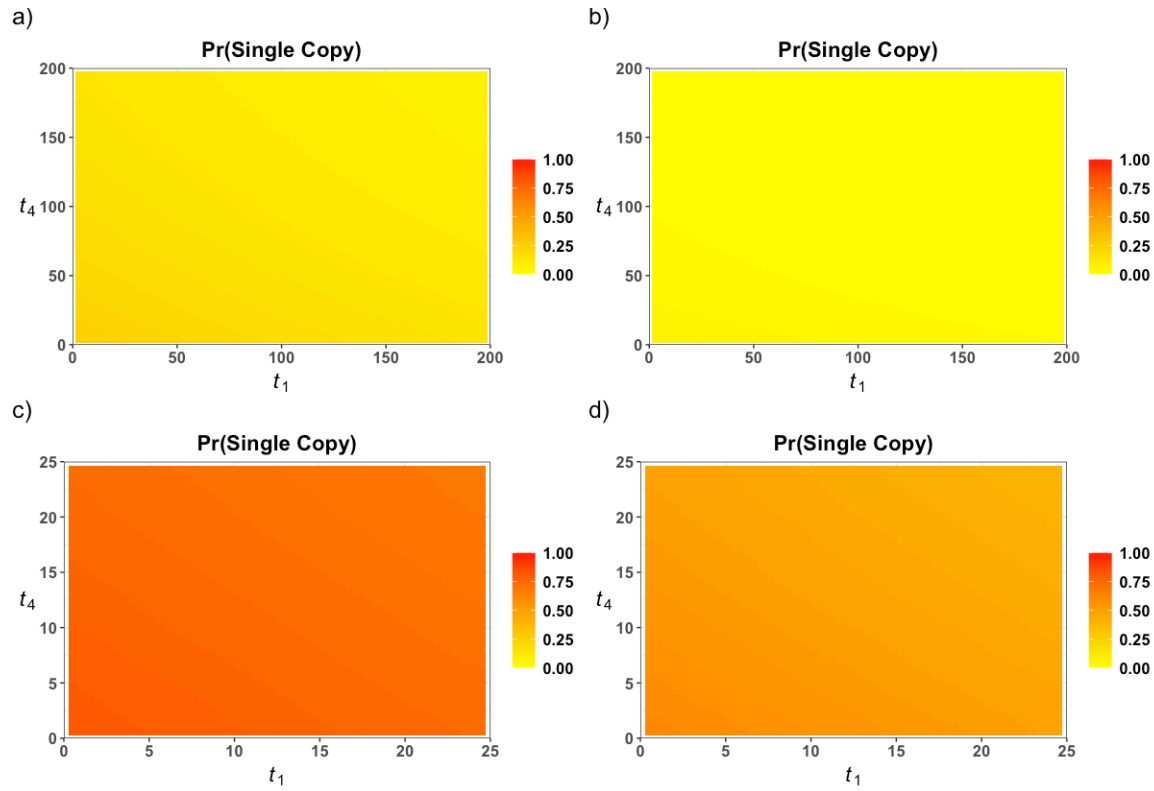


Fig S9. Probabilities of single-copy genes. a) The probability of single-copy genes with reasonable rates of duplication and loss ($\lambda=0.002$, $\mu=0.002$). Here, branch length $t_3 = 198.9$. Branch length t_1 varies from 0.0001 to 200, while branch length t_4 varies from 0.0001 to 198.8; branch length t_2 is constrained such that the sum of t_2 and t_4 equals t_3 . b) The probability of single-copy genes with high rates of duplication and loss ($\lambda=0.005$, $\mu=0.005$). Branch lengths are the same as in panel a. c) The probability of single-copy genes with reasonable rates of duplication and loss ($\lambda=0.002$, $\mu=0.002$). Here, branch length $t_3 = 25$. Branch length t_1 varies from 0.0001 to 25, while branch length t_4 varies from 0.0001 to 24.9; branch length t_2 is constrained such that the sum of t_2 and t_4 equals t_3 . d) The probability of single copy genes with high rates of duplication and loss ($\lambda=0.005$, $\mu=0.005$). Branch lengths are the same as in panel c.

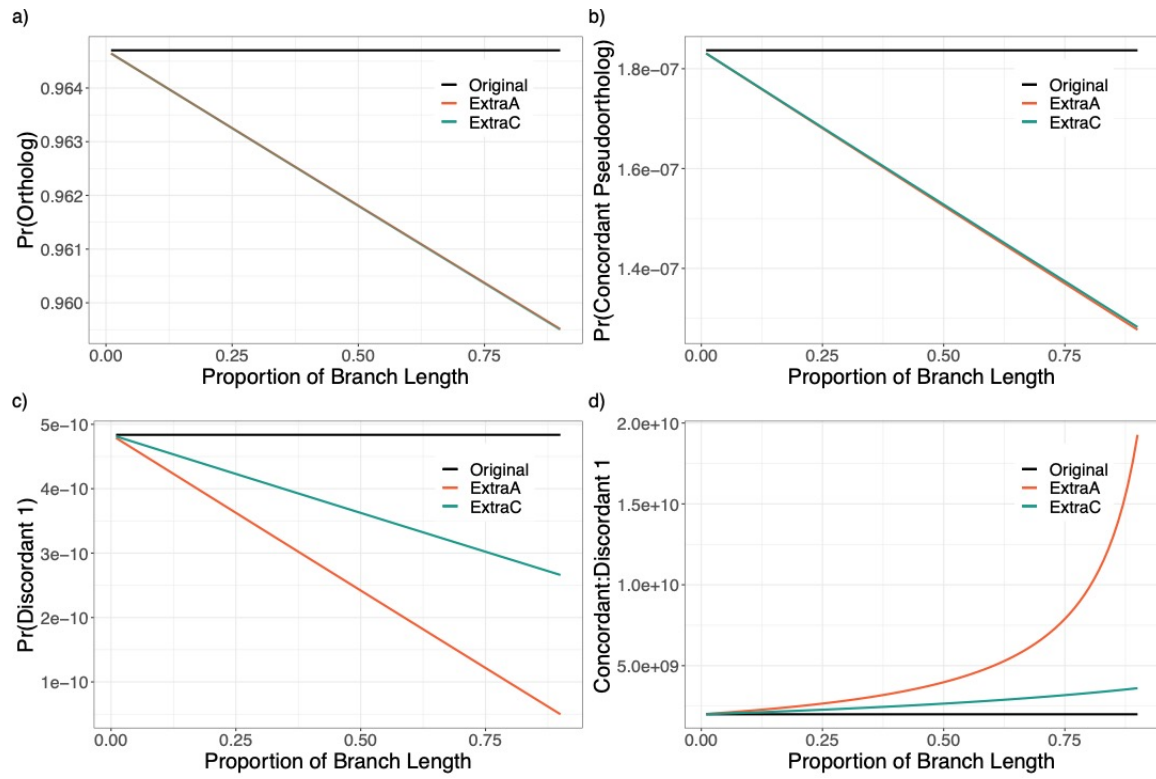


Fig S10. Results on larger trees. The original scenario includes a single branch per species. Here ExtraA includes an extra branch sister to Species A and ExtraC includes an extra branch sister to species C. The parameters for all panels are: $\mu = 0.004$, $\lambda = 0.002$, $t_1 = 1$, $t_2 = 1$, $t_3 = 2$, $t_4/t_5 = 1$. The length of the added branch is proportional to branch t_4 . a) The probability of orthologs. b) The probability of concordant pseudoorthologs. c) The probability of discordant pseudoortholog 1. d) The ratio of the concordant topologies to the probability of discordant pseudoortholog 1.

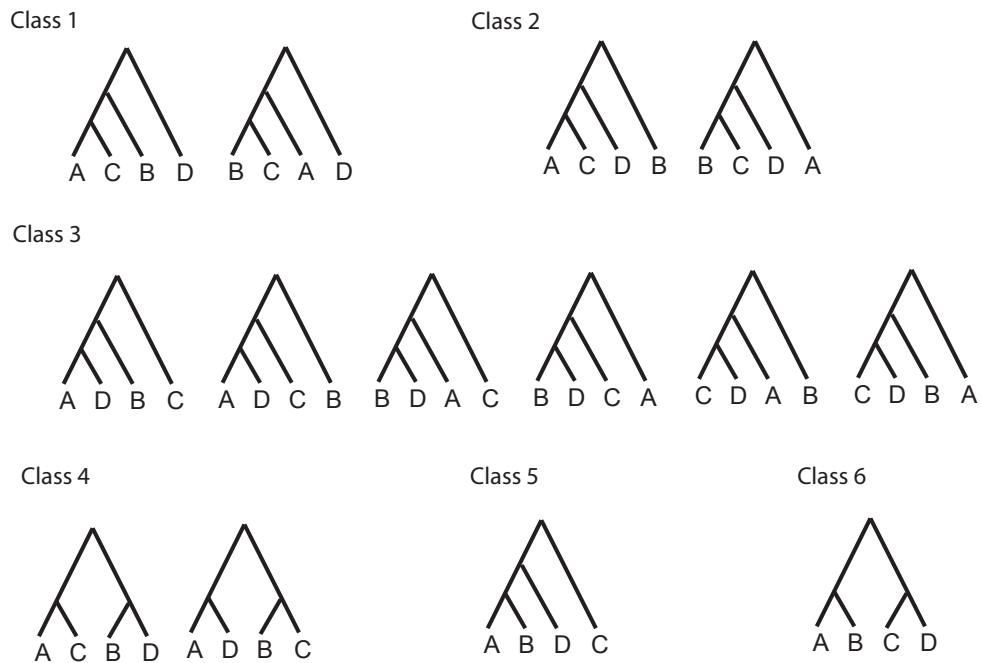


Fig S11. Discordant Topologies under the four-taxon case. Class labels correspond to those used for plotting in Figure S11.

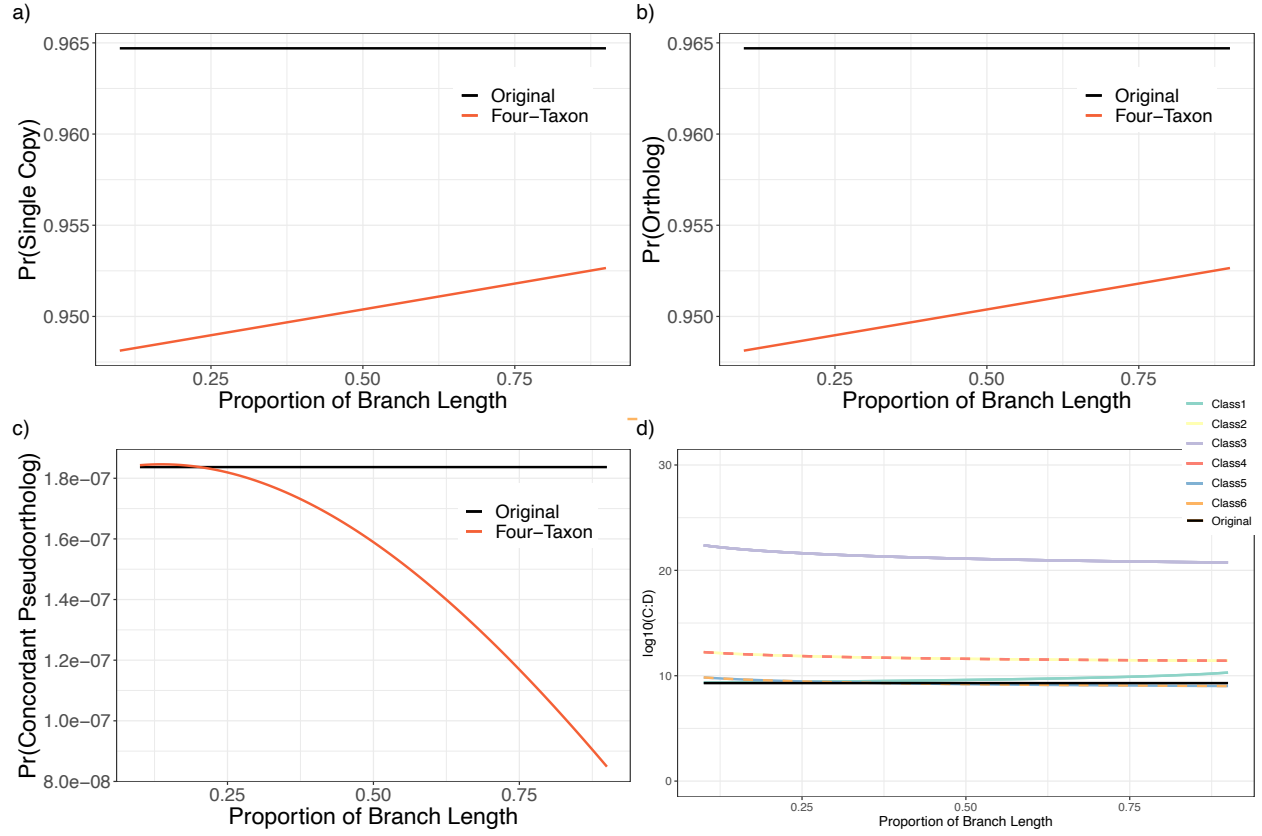


Fig S12. Results on larger trees. The original scenario includes a single branch per species. The four-taxon case includes an additional species 'D' sister to A, B, and C. The parameters for all panels are: $\mu = 0.004$, $\lambda = 0.002$, $t_1 = 1$, $t_2 = 1$, $t_3 = 2$, $t_4/t_5 = 1$ in the original case. For the four-taxon case, the length of branch t_1 was divided into branch t_{1a} and t_{1b} , with the proportion of the branch allocated to t_{1a} ranging from 0.01 to 0.9, and the length of branch t_6 was set such that the tree was ultrametric. a) The probability of single-copy genes. b) The probability of orthologs. c) The probability of concordant pseudoorthologs. d) The log of the ratio of the concordant topologies to each type of discordant pseudoortholog (classes shown in Figure S11; all discordant pseudoorthologs within each class have equal probabilities).

Table S1. Priors used in SimPhy simulations.

	<i>Minimum</i>	<i>Maximum</i>
μ	0.0001	0.005
λ	0.0001	0.005
t_1	0.01	200
t_2	0.01	200
t_3	0.01	200

Table S2. The parameters that maximize the relative probabilities of pseudoorthologs, maximize the relative probability of discordant pseudoorthologs, and minimize the ratio of concordant to discordant topologies. Discordant probabilities and ratios refer only to discordant pseudoortholog 1, but the probabilities of the two discordant pseudoorthologs are equal. SC=Single copy.

	<i>Pseudoorthologs</i>	<i>Discordant</i>	<i>Concordant: Discordant</i>
μ	0.0049	0.0050	0.0050
λ	0.0050	0.0050	0.0049
t_1	199.8807	199.9939	193.4689
t_2	0.5893	1.1028	0.2272
t_3	199.8666	199.4551	199.8339
t_4	199.2773	198.3523	199.6066
t_5	199.2773	198.3523	199.6066
	Pr(Pseudoorthologs SC) = 0.285	Pr(Discordant Pseudo SC) = 0.095	Concordant: Discordant = 8.53
	Pr(Orthologs SC) = 0.715	Pr(Concordant Pseudo SC) = 0.099	Pr(Discordant Pseudo SC) = 0.095
		Pr(Orthologs SC) = 0.711	Pr(Concordant Pseudo SC) = 0.096
			Pr(Orthologs SC) = 0.714

Table S3. Bounds on parameters used when performing optimization.

	<i>Minimum</i>	<i>Maximum</i>
μ	0.0001	0.005
λ	0.0001	0.005
t_1	0.0001	200
t_2	0.0001	200
t_3	0.0001	200

Table S4. The parameters that maximize the absolute probabilities of pseudoorthologs and the probability of discordant pseudoorthologs.

	<i>Pseudoorthologs</i>	<i>Discordant</i>
μ	0.0049	0.0043
λ	0.0031	0.0024
t_1	136.5054	152.1832
t_2	135.8794	0.1056
t_3	136.7011	150.6176
t_4	0.8217329	150.512
t_5	0.8217329	150.512
	Pr(Pseudoorthologs) = 0.0082	Pr(Discordant) = 0.0029
	Pr(Orthologs) = 0.0779	Pr(Concordant) = 0.0014
		Pr(Orthologs) = 0.0362

Table S5. The results of the comparison of simulations with WGDs to calculations under our model. All probabilities except the probability of single-copy genes ($\text{Pr}(\text{SC})$) are conditional on the sampled gene being single copy. $\text{Pr}(\text{O})$ is the probability of orthologs, $\text{Pr}(\text{C})$ is the probability of concordant pseudoorthologs, $\text{Pr}(\text{D1})$ is the probability of discordant pseudoortholog 1, and $\text{Pr}(\text{D2})$ is the probability of discordant pseudoortholog 2. The probabilities under WGD simulations are reported as '[WGD]'.

μ	λ	t_1/t_3	t_2	t_4/t_5	$\text{Pr}(\text{SC})$	$\text{Pr}(\text{SC} \text{WGD})$	$\text{Pr}(\text{O} \text{SC})$	$\text{Pr}(\text{O} \text{SC} \text{WGD})$	$\text{Pr}(\text{C} \text{SC})$	$\text{Pr}(\text{C} \text{SC} \text{WGD})$	$\text{Pr}(\text{D1} \text{SC})$	$\text{Pr}(\text{D1} \text{SC} \text{WGD})$	$\text{Pr}(\text{D2} \text{SC})$	$\text{Pr}(\text{D2} \text{SC} \text{WGD})$
0.002	0.002	50	25	25	0.511	0.100	0.999	0.959	0.001	0.038	0.000	0.001	0.000	0.002
0.002	0.002	100	50	50	0.276	0.104	0.992	0.921	0.007	0.064	0.000	0.005	0.000	0.010
0.002	0.002	200	1	199	0.072	0.047	0.961	0.837	0.013	0.061	0.013	0.053	0.013	0.049
0.005	0.005	50	25	25	0.207	0.094	0.986	0.902	0.012	0.087	0.001	0.006	0.001	0.004
0.005	0.005	100	50	50	0.059	0.048	0.927	0.819	0.062	0.165	0.006	0.006	0.006	0.010
0.005	0.005	200	1	199	0.006	0.007	0.709	0.691	0.099	0.059	0.096	0.147	0.096	0.103